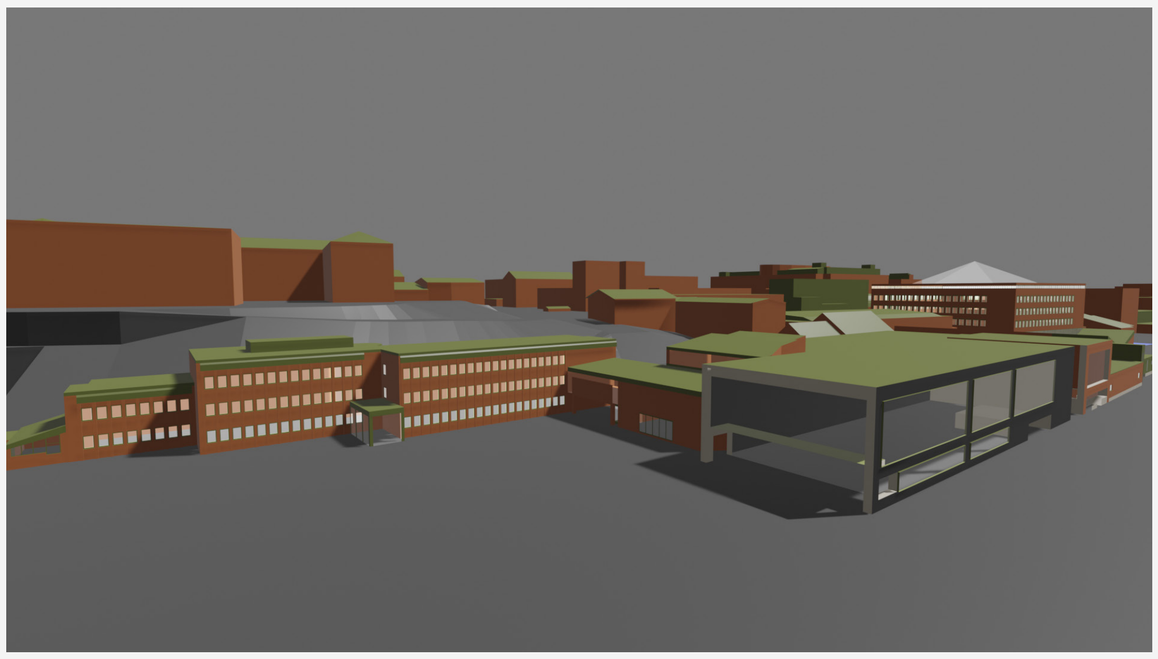
Digital Radio Twin of Chalmers

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Cellular communication has evolved from basic short distance voice transmission to global exchange of diverse forms of information. The next generation of cellular communication is 6G and for implementation, optimal configurations of the channel is sought after. These configurations are tested by evaluating data of ray tracing on a digital model to be optimized. The project's objective was to enable such a testing procedure for development and implementation of 6G.

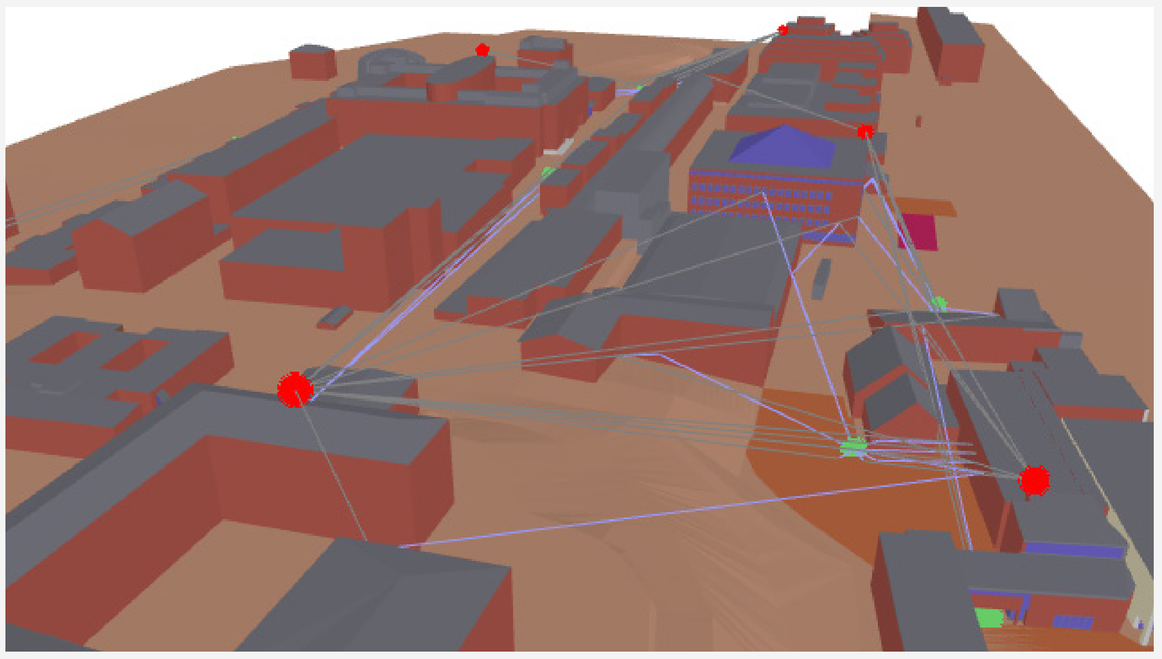


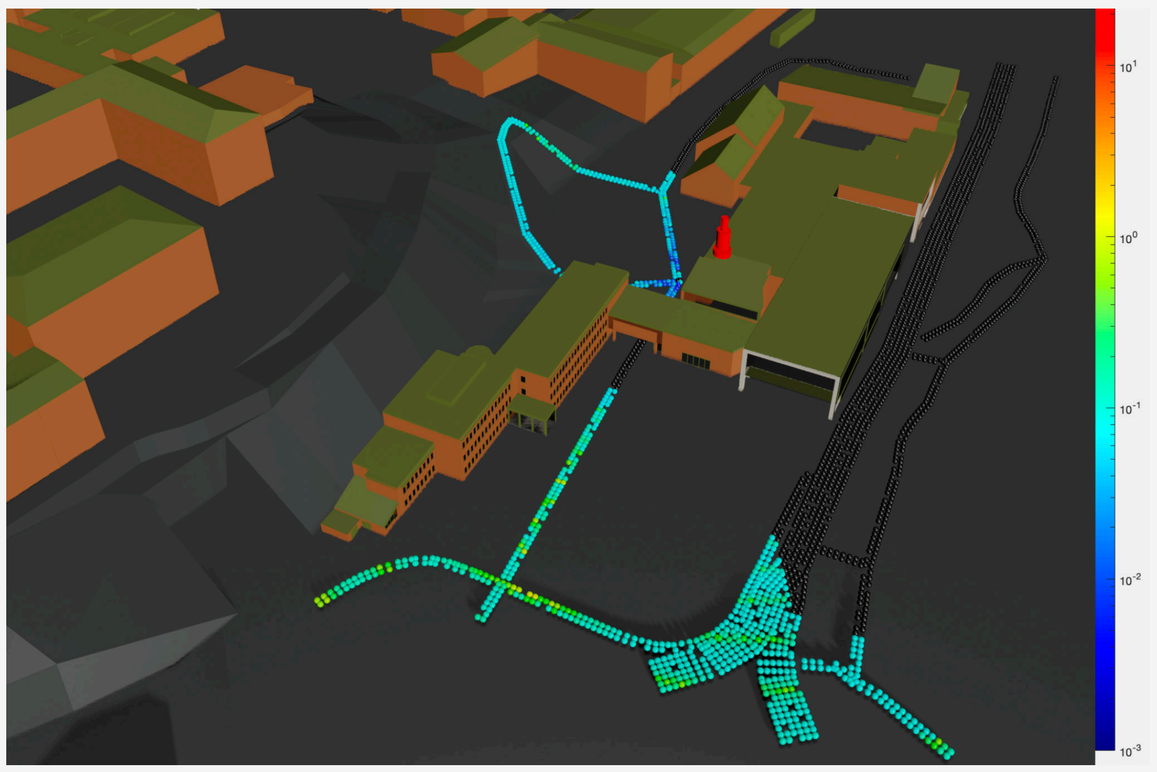
1 Designing the Digital Twin

The Digital Twin of Chalmers was created in Blender with add-ons and manual adjustments to more accurately reflect real world Chalmers. Structures and elevation was adjusted to reflect reality. Once the model had been created it was used for data generation through ray tracing.

2 Ray tracing & Data Generation

The ray tracing was done in python through NVIDIAs open source library Sionna. It enables simulation of signal propagation for testing and data generation. By placing a transmitter and receivers and deciding other channel parameters properties, a simulation can be executed and data extracted. The generated data was then evaluated through computations in MATLAB and lastly analyzed





3 Communication KPIs

The simulation data from Sionna was used to evaluate system performance by calculating Position Error Bound (PEB), Latency, and Capacity in MATLAB. These metrics served as Key Performance Indicators (KPIs), helping to identify trends and dependencies. Results showed that while distance affected PEB, shorter distances did not always lead to lower errors, as environmental complexity also played a role. Latency was mainly influenced by the number of transmitted bits, and Capacity increased with higher transmission power.